

Lipid Membranes as Active Interfaces: From Molecular Organization to Cellular Mechanics

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Lipid membranes are not merely passive boundaries but active interfaces that organize molecules and regulate the structure and mechanics of cell-like systems. In this talk, I will discuss how membrane-mediated interactions can be exploited to control molecular organization across different length scales. First, I will show how DNA origami nanostructures assembled on lipid membranes generate tunable mechanical properties in synthetic cells, enabling independent control of membrane bending and stretching. I will then discuss how membrane confinement and interfacial interactions promote structural reorganization of proteins, including β -sheet formation in cell-sized droplets. Together, these studies demonstrate how lipid interfaces couple molecular organization to the emergent properties of artificial cells.